

CS 3300: Intro to software Engineering

Mid-term Project Presentation
Software Development Plan Briefing

Project Title: Vibe Vision
Date: 2/26/26

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Syndeed Ayman



Engineers

I do struggle to comprehend

TEAM

Team Members and Roles

- Team A
 - Joshua – Github Maintainer + Kotlin dev
 - Lydia – Scrum master + Kotlin dev
- Team B
 - Syndeed – ML dev
 - Jared – Interfacing developer + ML dev

PROBLEM STATEMENT

- Some software applications rely on understanding user context to provide meaningful, adaptive experiences. There is a need for an accessible, on-device method to infer basic emotions – such as happiness, sadness, anger, or neutral – directly from camera input so that applications can modify their behavior without relying on external services or compromising user privacy
- Additionally, many engineering students fail to properly detect body language and facial emotions, leading to poor social performance and stress. There is a need for a system that helps them understand human emotion better.

PROJECT INTRODUCTION

- Project Title: **Vibe Vision**
- Project Description:
 - We plan to develop an Android application that builds upon a pre-existing machine learning model to determine whether the user appears happy, sad, angry, neutral based on an image captured through the device's camera. The app will process facial expressions in real time, classify emotional states using a trained image-based ML classifier, and provide feedback within the user interface.
 - Python interface with Kotlin through TensorFlow

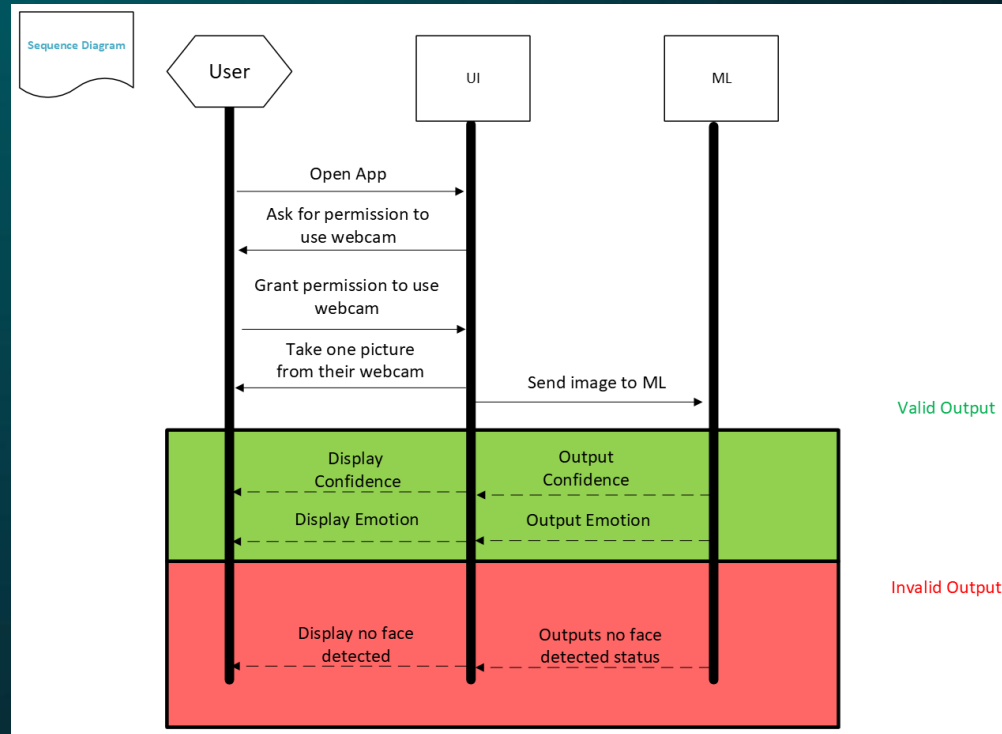
SOFTWARE ENGINEERING PRINCIPLES

- **Agile** because requirements may evolve and we want continuous feedback (reevaluation).
- **Why Agile?**
 - Requirements may change
 - Fast development needed
 - Team feedback is important
 - Team collaboration is strong
- **Why Scrum?**
 - Clear structure
 - Defined roles
 - Short development cycles (Sprints)
 - Easy progress tracking

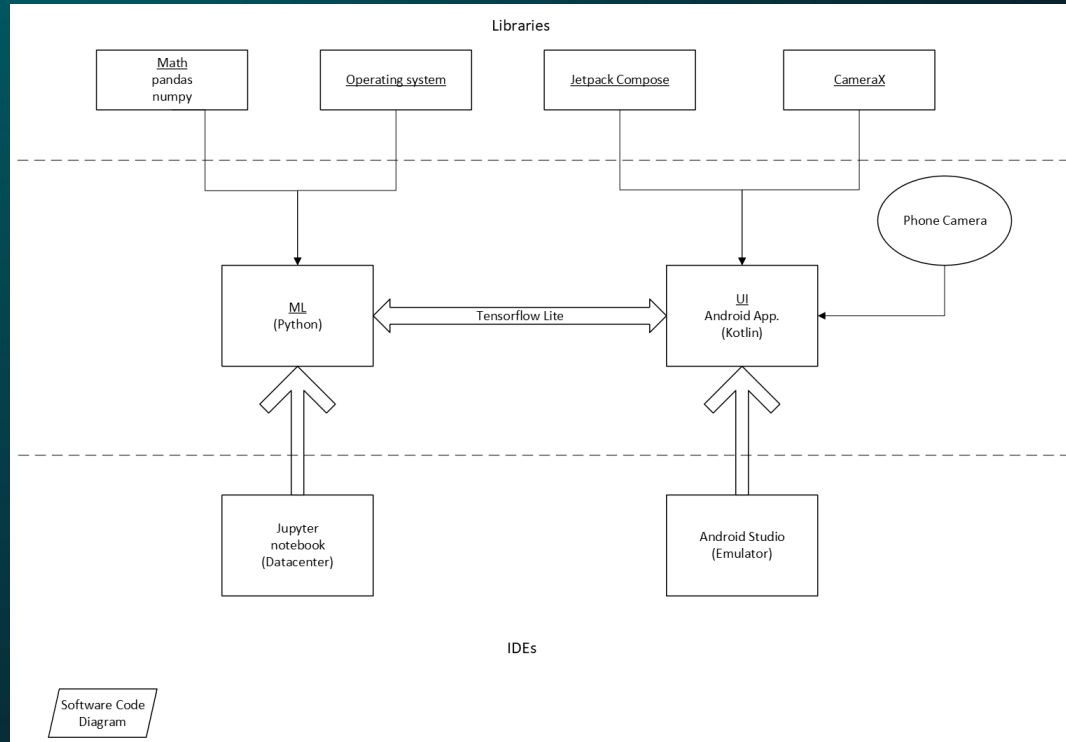
SOFTWARE ENGINEERING PRINCIPLES CONT.

- **Pair Programming**
 - Two developers work together
 - Higher quality code
 - Knowledge sharing
- **More than one method**
 - We'll use Agile method and documentation standards from Plan-driven

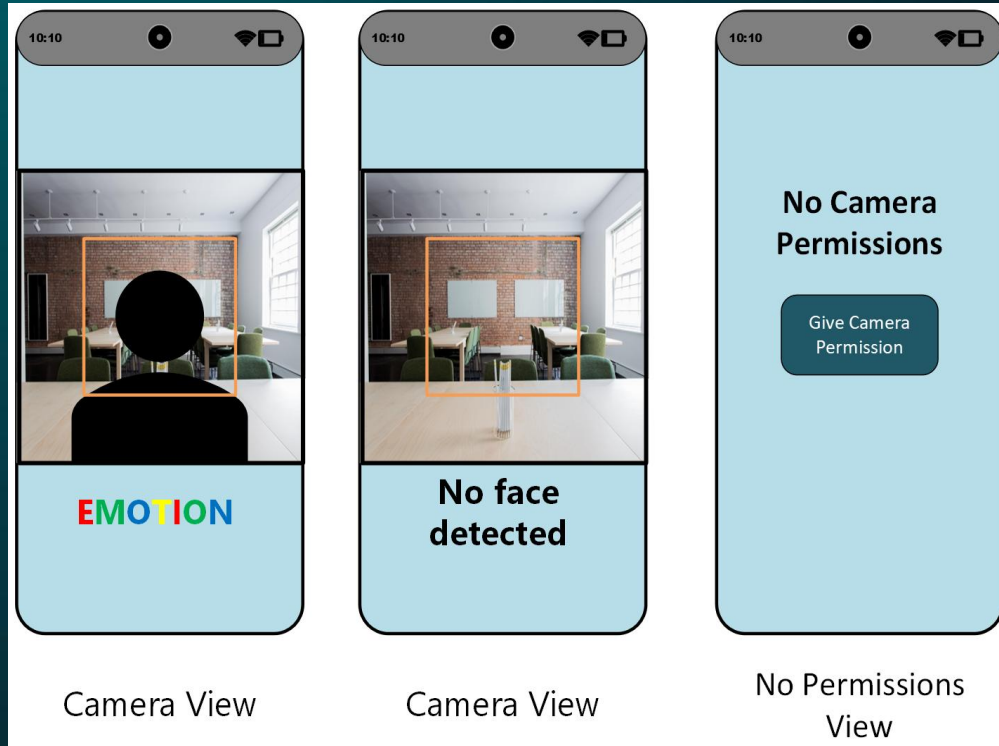
PROJECT ARCHITECTURE DIAGRAM



PROJECT SOFTWARE DESIGN DIAGRAM



PROJECT PERSPECTIVE MODEL DIAGRAM



PROJECT FUNCTIONAL REQUIREMENTS

1	Module: Machine Learning
1.1	Samples image data
1.1.1	Uses images that must be 720p
1.1.2	Image is from computer webcam
1.1.3	Processes a single frame at a time
1.2	Input Data: Detecting human faces
1.2.1	Uses input images
1.2.1	Within 90% accuracy to detect human face in dataset
1.2.2	Within 70% accuracy to detect human face in app
1.2.3	Returns "no face detected" status
1.3	Detects human expression
1.3.1	Within 60% accuracy to classify human expression in
1.3.2	Within 40% accuracy to classify human expression in
1.3.3	Returns confidence of the model
1.3.4	Throws error if not recognized emotion {happy, sad, a
1.3.5	Reapproach the training principles for model
1.3.5.1	Re-asses dataset valid for mobile cameras
1.3.5.2	Preprocess training data - look for CSV alternative
1.3.5.3	ReUse FER2013 dataset
1.3.6	Not using FER2025 and HyperAI datasets

2	App Functionality
2.1	Event: App opens
2.1.1	Opens interface
2.1.2	Asks permission to use webcam
2.1.3	Opens webcam
2.2	Event: App closes
2.2.1	Turns off webcam
2.3	Interface: Machine Learning & App
2.3.1	App sends image data to ML
2.3.2	App receives ML computed results
2.4	UI: App displays results
2.4.1	Displays text that states emotion category results
2.4.2	Displays whether faces were detected
2.4.3	Displays confidence of model
2.5	Runs on desktop Kotlin environment (JVM), not Android

PROJECT NON-FUNCTIONAL REQUIREMENTS

3.1	Safety	6	App things
3.1.1	The app never connects to the local network	6.1	Be ran on local phone
3.1.2	App does not store user images	6.2	Uses Kotlin
3.2	Robustness: No handling of edge cases	6.3	Color coding based on confidence of model/emotion
3.2.1	Doesn't need to handle unclear pictures	6.4	User ability to flip from webcam to front cam
3.2.2	No more than one face in frame at a time	6.5	Display different message for no confidence
3.3	Logging	7	Additional information
3.3.1	No errors are logged	7.1	Returns confidence threshold
3.4	Accessibility	8	Robustness: Handles edge cases
3.4.1	UI text is readable at 100% scaling, colors contrast	8.1	Suboptimal conditions
The System May... (Probably Rev2 stuff)		8.1.1	Behavior under poor lighting
4	Speed	8.1.2	Behavior when face is partially occluded
4.1	Less than a minute to compute emotion	8.1.3	Behavior when multiple faces appear
4.2	Less than thirty seconds to open app	8.1.4	Behavior when camera is blurry
5	Image processing	8.2	Logging
5.1	Uses buffer for smoother processing	8.2.1	Logging inference times
		8.2.2	Logging recognition failures
		8.2.3	User problem logging

PROJECT NON-FUNCTIONAL REQUIREMENTS

9	System Requirements
9.1	The system must run on a desktop JVM environment (Kotlin).
9.2	The system must have access to a webcam capable of 720p or higher.
9.3	The system must run entirely offline.
9.4	The ML model must execute locally on the same machine.

PROJECT SCHEDULE

	Start Date	End Date	8-Feb	16-Feb	23-Feb	2-Mar	9-Mar	16-Mar	23-Mar	30-Mar	6-Apr	13-Apr	20-Apr	27-Apr	4-May
PLANNING															
Establish team	20-Jan	1-Feb													
Initial Requirements	2-Feb	8-Feb													
Architecture diagram	2-Feb	8-Feb													
Software design diagram	2-Feb	8-Feb													
MID-TERM presentation															
20-Jan	19-Feb														
SPRINT #1 - initial logic															
IT 0 - Establish ML & App Workspace															
Team A: ML Revamp															
ML Reformatting For Phone Cameras															
Training Data for Phone Cameras															
Train the ML Model															
IT 1 - Establishing Interfaces															
TEAM A: Interface ML with Kotlin	8-Feb	22-Feb													
Send simplified data from ML to Kotlin	8-Feb	15-Feb													
Unit Test	15-Feb	22-Feb													
TEAM B: Foundational Kotlin development	8-Feb	22-Feb													
Send example image/bitmap to ML	8-Feb	15-Feb													
Unit Test	15-Feb	22-Feb													

2/26/2026

Lydia Easter

PROJECT SCHEDULE

	Start Date	End Date	8-Feb	16-Feb	23-Feb	2-Mar	9-Mar	16-Mar	23-Mar	30-Mar	6-Apr	13-Apr	20-Apr	27-Apr	4-May
IT 2 - Adding complexity to interfacing	23-Feb	8-Mar													
TEAM A: Interface ML with Kotlin	23-Feb	8-Mar													
Send full package of data from ML to Kotlin	23-Feb	1-Mar													
Unit Test	2-Mar	8-Mar													
TEAM B: Foundational Kotlin development	23-Feb	8-Mar													
Open camera and take picture	23-Feb	1-Mar													
Unit Test	2-Mar	8-Mar													
IT 3 - establish MVP	8-Mar	22-Mar													
TEAM A: Get Kotlin widget to show ML data	8-Mar	22-Mar													
Test ML capabilities on images from Kotlin	8-Mar	22-Mar													
TEAM B: Ensure images send	8-Mar	22-Mar													
Pictures that are taken with camera get uploaded	8-Mar	15-Mar													
Test to ensure the ML image received matches	15-Mar	22-Mar													
Integration Testing - MVP is established															
Time buffer for Spring #1															
Planning for Sprint #2															
Unit Test															

PROJECT CONFIGURATION MANAGEMENT

- Environment stability: Joshua manually follows the same procedure for each member of the team. Works for small teams
- Github: used to organize, track, and control changes to the system so it stays stable, consistent, and reliable over time.
- Changelog for versions of code, maintained list of improvements, fixes, and bugs

PROJECT CONFIGURATION MANAGEMENT

- Hardware Configuration:
 - GPU: A100 GPU (Google Collab)
 - GPU RAM: 80GB
 - System RAM: 168GB
- Model configuration:
 - Model architecture: Sequential Convolutional Neural Network
 - Input format: 128*128*1 (greyscale)
 - Loss function: sparse_categorical_crossentropy
 - Optimizer: Adam
 - Batch Size: 32 of 128x128 images

PROJECT SOFTWARE DEVELOPMENT

- Open-source utilities: libraries
 - Python: TensorFlow, Keras, NumPy, Matplotlib
 - Kotlin: Jetpack Compose, CameraX
- Java Development Kit (JDK) version: 17
- Software Development Kit (SDK) version: android 36.1
- Python 3.12.7
- Gradle 9.2.1
- Java 11
- Android Studio
- Jupyter

PROJECT VERIFICATION

- How do you plan to verify (test) your requirements?
 - Using the app with different test cases including different people, emotions, and lack thereof
 - Check for the deletion of User Data after use
 - Run many tests to get accuracy data
 - Check Confidence values

ETHICS

- What are your Ethical concerns?
 - Saving of User data
 - Lack of use for Diverse Skin tones, Lighting, and Occlusions
 - Utilization of Data
- How will you avoid ethical pitfalls?
 - Data Deletion
 - Diverse Data Set
 - No Cloud Service
 - Asking for Camera Permissions

RISK MANAGEMENT

Personal

- Snow days: Microsoft Teams meetings if needed
- Teammate MIA: Confront teammate in class

Hardware

- Low End Devices: Android SDK minimum version 24
- File Loss: Using GitHub Repository

Software

- User doesn't accept permissions: Don't access camera and display button to allow permissions
- Integration Problems: start integration early between model and app in emulator, set time for physical hardware integration

PROJECT COMPLETION

- What is the final output of your project? What is your planned final product? How are you defining DONE?
 - A android app that detects emotions based on real time camera input. A minimal viable product
- Do you plan to license your code?
 - Yes, using the MIT license mainly to ensure our backs are covered if someone else uses our code.
- Is your code open-source?
 - Yes – MIT license
- What is the future for your project?
 - Implementing on hardware, increasing the accuracy, facial detection

The background is a dark teal color with a complex pattern of thin, light blue lines forming a network or web-like structure. Scattered throughout are numerous small, out-of-focus circles in shades of red, orange, and yellow, creating a bokeh effect. A bright, multi-colored light source is visible in the lower center, casting a soft glow.

Any Questions?